

C3 Stoichiometry

C3.1 Formulas

Stoichiometry is the amounts of substances involved in chemical reactions. It helps us understand how much of each reactant we need and how much product will be formed. By using formulae and balanced chemical equations, we find relationships between quantities of reactants and products.

State the formulas of the elements and compounds named in the subject content

- This means know some common chemicals e.g. $C_6H_{12}O_6$ is glucose, H_2SO_4 is sulfuric acid, NaOH is sodium hydroxide etc. *Try to learn them as you meet them in the course.*

Define the molecular formula of a compound as the number and type of atoms in one molecule & deduce the formula of a simple molecular compound from the relative numbers of atoms present in a model or a diagrammatic representation

- The molecular formula of a compound shows the actual number of each type of atom present in a molecule of that compound. It provides the types and quantities of atoms without indicating how they are arranged or bonded.

Examples:

Water (H_2O): Contains 2 hydrogen atoms and 1 oxygen atom.

Carbon Dioxide (CO_2): Contains 1 carbon atom and 2 oxygen atoms.

Glucose ($C_6H_{12}O_6$): Contains 6 carbon atoms, 12 hydrogen atoms, and 6 oxygen atoms.

Construct word equations to show how reactants form products

- Identify the substances that are reacting together. These are your reactants.
- Identify what substances are formed as a result of the reaction. These are your products.
- Write the equation using the following format: Reactants $\rightarrow$ Products
- Write the names of the reactants on the left side, followed by an arrow ($\rightarrow$), and then write the names of the products on the right side.
- Use "+" for multiple reactants/products: e.g. reactant 1 + reactant 2 $\rightarrow$ product 1 + product 2

Example: Combustion of methane

Reactants: methane and oxygen Products: carbon dioxide and water

Word equation: methane + oxygen $\rightarrow$ carbon dioxide + water

Balance and interpret simple symbol equations, including state symbols

- To balance symbol equations, adjust coefficients (the numbers in front of the formulas) to ensure the same number of each type of atom on both sides. State symbols indicate the physical state: (s) for solid, (l) for liquid, (g) for gas, and (aq) for aqueous.

Example: hydrogen gas burns in oxygen gas to make liquid water

Unbalanced Equation: $H_2 + O_2 \rightarrow H_2O$

Count Atoms: on the left of $\rightarrow$: 2 H, 2 O, on the right of $\rightarrow$: 2 H, 1 O

Adjust Coefficients: $2H_2 + O_2 \rightarrow 2H_2O$

($2H_2$ means H_2 and H_2 i.e. 4 x H atoms)

($2H_2O$ means H_2O and H_2O i.e. 4 x H atoms and 2 x O atoms)

Add State Symbols: $2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$

Supplement

Deduce the formula of an ionic compound from the relative numbers of the ions present in a model or a diagrammatic representation or from the charges on the ions

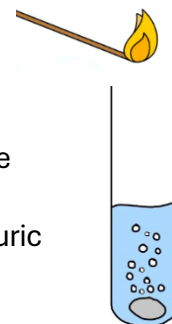
- Identify the positive ions (cations) and negative ions (anions) present in the compound. Some common cations include metals (e.g. Na^+ , Ca^{2+} , Al^{3+}) and common anions include nonmetals and polyatomic ions (e.g. Cl^- , O^{2-} , SO_4^{2-} , OH^- and CO_3^{2-}).
- Note the charge on *each ion*. This is important because the *total* positive charge must balance the total negative charge.
- Balance the charges: use the charges to work out the ratio of cations to anions needed to be electrically neutral:
 - If the cation has a charge of +1 and the anion has a charge of -1, they will combine in a 1:1 ratio (e.g. NaCl).
 - If the cation has a charge of +2 and the anion has a charge of -1, two anions are needed for each cation (e.g. CaCl_2).
- Write the formula by combining the cations and anions in the ratio determined to write the formula:
 - Use subscripts to indicate the number of each ion needed. If only one ion is present, no subscript is written. Example: If the ratio is 1:2 for Mg^{2+} and Cl^- , the formula is MgCl_2 .

Example: Na^+ and SO_4^{2-} ions

Na^+ has a charge of 1+ and SO_4^{2-} has a charge of 2-

To balance the 2- charge from SO_4^{2-} you need two of the Na^+ ions

The formula of the ionic compound sodium sulfate is Na_2SO_4



Construct symbol equations with state symbols

- This is more difficult as you need to know the reactants, and what they react to make (the products).
E.g. you are expected to remember that a piece of magnesium ribbon will react with sulfuric acid solution to make magnesium sulfate solution and hydrogen gas (which burns with a squeaky pop).
 - Identify reactants (magnesium and sulfuric acid) and products (magnesium sulfate and hydrogen)
 - Write the correct chemical formulas for all reactants and products. Use an arrow ($\rightarrow$) to separate reactants from products. $\text{Mg} + \text{H}_2\text{SO}_4 \rightarrow \text{MgSO}_4 + \text{H}_2$
 - Balance the equation (if necessary).
 - Include state symbols next to each chemical formula to indicate physical state: (s), (l), (g) or (aq)
 $\text{Mg(s)} + \text{H}_2\text{SO}_4\text{(aq)} \rightarrow \text{MgSO}_4\text{(aq)} + \text{H}_2\text{(g)}$ Remember (l) = liquid (aq) = solution

Construct ionic equations with state symbols (This is hard – don't panic too much!)

- Write out the full balanced equation e.g. $2\text{KI(aq)} + \text{Cl}_2\text{(aq)} \rightarrow 2\text{KCl(aq)} + \text{I}_2\text{(aq)}$
- Replace ionic compounds* with their ions (*compounds of metal cations + non-metal anions)
 $2\text{K}^+\text{(aq)} + 2\text{I}^-\text{(aq)} + \text{Cl}_2\text{(aq)} \rightarrow 2\text{K}^+ + 2\text{Cl}^-\text{(aq)} + \text{I}_2\text{(aq)}$
- Remove any ions that appear on each side of the $\rightarrow$
 $2\text{I}^-\text{(aq)} + \text{Cl}_2\text{(aq)} \rightarrow 2\text{Cl}^-\text{(aq)} + \text{I}_2\text{(aq)}$